

Release Notes for AT32A423 Firmware Library

V2.0.5-2026/06/02

1. CMSIS
 - system_at32a423.c
 - ◆ Updated the System_init() to avoid possible clock failure during switching
2. Middlewares
 - I2C
 - ◆ Corrected these macro definitions:
I2C_MEM_ADDR_WIDIH_8 is changed to I2C_MEM_ADDR_WIDTH_8;
I2C_MEM_ADDR_WIDIH_16 is changed to I2C_MEM_ADDR_WIDTH_16
 - USB
 - ◆ Updated the usbh_enum_handler() in host mode to enhance compatibility: add a delay of 10ms after SET_ADDRES operation to ensure that the device address is taken into account before starting subsequent configuration and operation.
 - ◆ Updated “msc class” by adding a step to handle the test_unit() and the start_stop(), in order to enhance compatibility
 - ◆ Updated “msc class” to solve the problem of encountering a compiling warning issued in the uhost_process_handler() when the USBH_DEBUG_ENABLE is not activated
3. Drivers
 - PWC
 - ◆ __force_stores() is replaced with “__ISB()” and “__DSB()” to adapt to different compilers
 - CRM
 - ◆ Updated the processes of the crm_reset(), the crm_hick_sclk_frequency_select() and the crm_hick_divider_select() to avoid possible clock failure during switching
 - EXINT
 - ◆ Added a new EXINT_TRIGGER_NONE_EDGE to adapt to software trigger without edge configuration requirements
4. Demos
 - I2C
 - ◆ Corrected I2C demo macro definitions:
I2C_MEM_ADDR_WIDIH_8 is changed to I2C_MEM_ADDR_WIDTH_8;
I2C_MEM_ADDR_WIDIH_16 is changed to I2C_MEM_ADDR_WIDTH_16
5. Updated some notes

V2.0.4-2025/11/24

1. CMSIS
 - at32a423.h
 - ◆ Removed unnecessary base address macros
2. Drivers
 - ADC
 - ◆ Removed unnecessary bit name “spt_bit.cspt18” and the register csts_bit
 - CRM
 - ◆ Changed the bit name from adc14rst to tmr14rst
 - ◆ Removed excess definitions in the CRM_xxx_PERIPH_LOWPOWER
 - TMR
 - ◆ Updated these functions: tmr_pwm_input_config(), tmr_encoder_mode_config() and tmr_output_channel_config()
The channel must be disabled before configuring channel mode, otherwise it fails to switch between channel modes
 - Updated some notes
3. Middlewares
 - USB
 - ◆ Updated “keyboard class” to add a new member “send_state” in the keyboard_type structure to record send state, and updated the flow control handling in the usb_keyboard_class_send_report, so as to avoid data error
 - ◆ Updated “audio class” to change the audio sampling frequency request configuration, so as to solve host compatibility issue
4. Demos
 - ADC
 - ◆ Updated ADC initialization process in all demos to avoid errors occurring under extreme circumstances
 - ◆ Updated DMA-related demos to enable DMA circular mode during configuration to achieve circular trigger
 - ◆ Removed the software_trigger_repeat demo
 - USB
 - ◆ Updated the clock division configuration process in the usb_device demo to prevent USB clock division configuration error as a result of AHB division
 - ◆ Adjusted the procedure of configuring Deepsleep extra low-power mode in keyboard, and updated the latency method to achieve stabilization after wakeup
 - ◆ Adjusted the procedure of configuring Deepsleep extra low-power mode in mouse, and updated the latency method to achieve stabilization after wakeup
 - ◆ Updated the USB_FLASH_ADDR_OFFSET value in msc from 0x08005000 to 0x08008000, so as to solve compiling range overflow issue in different environments
 - ◆ Updated Jump APP function in the virtual_msc_iap demo to add compiling non-optimization settings, so as to avoid Jump failure arising from optimization settings
 - PWC

- ♦ Updated some demos to remove unnecessary steps for regulating voltage in the previous low-power mode entry procedure
- templates
 - ♦ Updated the “include” path in the eclipse_gcc to prevent compiling error due to incorrect parameters in the workspace_project_locations during multi-project import process
- Board
 - ♦ Updated the description of some parameter variables in board.c to solve warning issued when using a version of compiler
- Cortex-m4
 - ♦ Updated the description of latency judgement in the systick_interrupt from “greater than” to “greater than or equal to”, to avoid the problem “it takes one more 1ms than expected to toggle IO”
- AT32IDE and Eclipse projects
 - ♦ Removed “release” items and reserved “debug” items
- IAP
 - ♦ Updated all IAP_Programmer.exe
- Keil projects
 - ♦ Updated the IROM size configuration in some Keil projects

V2.0.3-2025/01/20

1. Updated some SPI demos.
2. Updated USB driver and some demos.
3. Updated IAP_Programmer.exe.
4. Updated LDO output level settings.
5. Updated some notes.

V2.0.2-2024/12/13

1. Updated core DSP library.
2. Updated CAN driver.
3. Added support for AT32IDE project in SLIB demo.
4. Updated USB driver.
5. Updated I2C eeprom demo and added multi-page write logic.
6. Optimized PWR driver and demo.
7. Updated some notes.

V2.0.1-2024/08/08

1. Updated IROM project configuration for the cmsis_dsp demo.
2. Updated configuration process of PWC low-power mode related demos.
3. Fixed AC6 compilation problems of some demos.
4. Updated some notes.

V2.0.0-2024/04/15

1. Initial release of AT32A423 series firmware library.